Assessment status: Preliminary. Reviewed and approved as Preliminary.

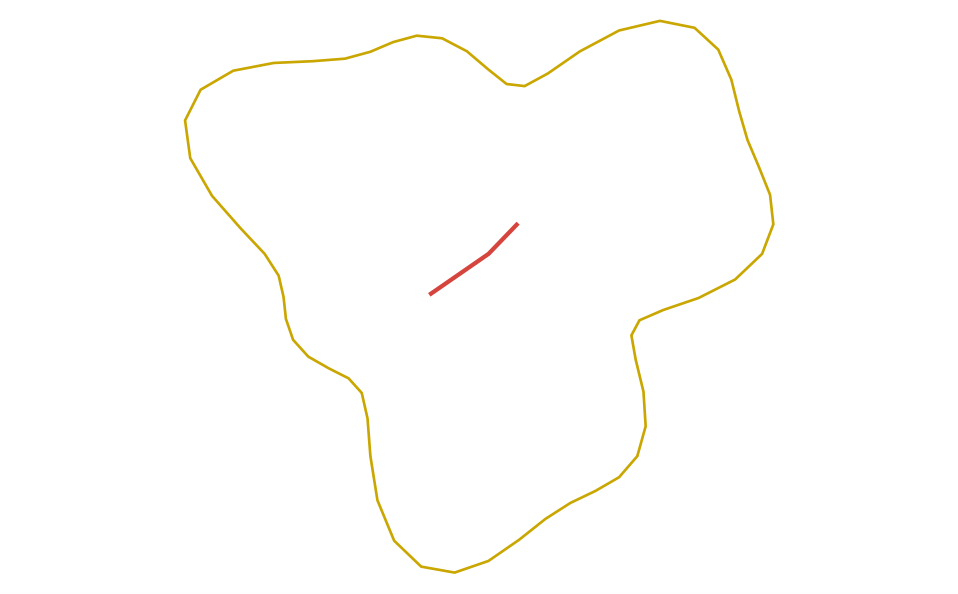
DEEP Detailed Assessment

Eastern Corn Belt Plains reference assessment

Mink Brook

Source	StreamCurves reference-curve development
Version / status	v6 · Preliminary
Predictor source	streamcat
Region match	Not resolved
Coordinates	43.68581234, -72.23671234
COMID / HUC8	9327042 / 01080104
Drainage area	12.43 km2
HR reach watershed area	11.98 km2
Reach length	1000.0 ft
Watershed basis	HR reach watershed (STAF site engine v0.5.0)
Desktop values	STAF site engine (HR reach watershed) 5, StreamCat lookup engine (NHDPlus V2 basin) 2
StreamCat reach	COMID 9327042 (this reach)
Content digest	sha256:bc2948a42062bb90030672afe31eb3271f563c790dee145c5cb8d7af8dfa42fd
Ecosystem Condition Index	0.45 to 0.82, condition not determinable, 4 of 20 functions not assessed
STAF function coverage	16 of 20 (4 documented exclusions)
Flagged transfers	none

Watershed



Map background unavailable

Function scores (0-15)

Assessment status: Preliminary. Reviewed and approved as Preliminary.

Function	Score	Condition
Catchment hydrology	12.0	F
Surface water storage		Not scored
Reach inflow	5.9	AR
Streamflow regime	8.4	AR
Low flow and baseflow dynamics	8.8	AR
High flow dynamics	9.5	AR
Floodplain connectivity	14.9	F
Hyporheic connectivity		Not scored
Channel evolution	14.9	F
Channel and floodplain dynamics	15.0	F
Sediment continuity		Not scored
Bed composition and bedform dynamics		Not scored
Light & thermal regime	7.3	AR
Carbon processing		Not scored
Habitat provision		Not scored
Watershed connectivity	12.3	F

Outcome sub-indices

Physical	Chemical	Biological	ECI
0.75	0.6	0.77	not assessed

Metric measurements & curve indices

Function	Metric	Value	Index	Source	Curve (source)
Catchment hydrology	Agriculture, crop and hay	3.0	0.97	desktop: STAF site engine v0.5.0 impervious (HR reach watershed, NLCD 2021)	EASI fixed criteria, Catchment hydrology (land-cover pressure)
Catchment hydrology	Impervious surface	12.75	0.64	desktop: StreamCat PCTIMP2019WS	EASI fixed criteria, Catchment hydrology (land-cover pressure)
Surface water storage	Wetland	0.4	ref. only	desktop: Computed (reference only)	StreamCurves reference-curve development
Reach inflow	Road density	3.0	0.39	desktop: STAF site engine v0.5.0 impervious (HR reach watershed, NLCD 2021)	EASI fixed criteria, Road-density proxy for concentrated inflow pressure
Streamflow regime	Base flow index	12.75	0.18	desktop: StreamCat PCTIMP2019WS	StreamCurves reference-curve development
Streamflow regime	Degree of regulation	0.4	0.94	desktop: Computed	EASI fixed criteria, Flow alteration (degree of regulation)
Low flow and baseflow dynamics	Bankfull width:depth ratio	7.5	0.99	field	StreamCurves reference-curve development

Assessment status: Preliminary. Reviewed and approved as Preliminary.

Function	Metric	Value	Index	Source	Curve (source)
Low flow and baseflow dynamics	Residual pool depth				StreamCurves reference-curve development
Low flow and baseflow dynamics	Base flow index	12.75	0.18	desktop: StreamCat PCTIMP2019WS	StreamCurves reference-curve development
High flow dynamics	Impervious surface	12.75	0.64	desktop: StreamCat PCTIMP2019WS	EASI fixed criteria, Catchment hydrology (land-cover pressure)
Floodplain connectivity	Bankfull width:depth ratio	7.5	0.99	field	StreamCurves reference-curve development
Floodplain connectivity	Bankfull height above channel				StreamCurves reference-curve development
Hyporheic connectivity	Sand + fines			field	StreamCurves reference-curve development
Hyporheic connectivity	Embeddedness				StreamCurves reference-curve development
Channel evolution	Bankfull width:depth ratio	7.5	0.99	field	StreamCurves reference-curve development
Channel evolution	Bankfull height above channel				StreamCurves reference-curve development
Channel and floodplain dynamics	Sinuosity	12.5	1.00	field	StreamCurves reference-curve development
Channel and floodplain dynamics	Bank angle				StreamCurves reference-curve development
Sediment continuity	Relative bed stability				StreamCurves reference-curve development
Sediment continuity	Sand + fines			field	StreamCurves reference-curve development
Bed composition and bedform dynamics	Substrate size				StreamCurves reference-curve development
Bed composition and bedform dynamics	Large wood volume				StreamCurves reference-curve development
Bed composition and bedform dynamics	Embeddedness				StreamCurves reference-curve development
Light & thermal regime	Canopy density, mid-channel	17.5	0.48	field	StreamCurves reference-curve development
Carbon processing	Riparian veg cover, woody				StreamCurves reference-curve development
Habitat provision	Fast-water / riffle habitat				StreamCurves reference-curve development
Habitat provision	Natural fish cover				StreamCurves reference-curve development
Watershed connectivity	Mapped dams within one mile	0.4	0.82	desktop: Computed	EASI fixed criteria, Nearby dam-proximity proxy

Assessment status: Preliminary. Reviewed and approved as Preliminary.

Scoring advisories

Wetland: engine-computed value shown as reference only: this curve was fitted on StreamCat predictors, so scoring it against an HR reach watershed value would mix training and serving sources

Sinuosity: value 12.5 is above the curve domain [1.0, 1.6272]; score clamped to the endpoint

Reference support

Reference curves are built from least-disturbed stations chosen by a fixed landscape-pressure screen. Where this ecoregion has too few, comparable stations of its Level II and then its Level I ecoregion are used, and a metric with no defensible pool is not scored. Landscape pressure metrics are scored on fixed criteria.

Metric	Scored against	Curve set
Agriculture, crop and hay	Fixed criteria, the same in every region	
Impervious surface	Fixed criteria, the same in every region	
Wetland	Reference curve from 10 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high	
Road density	Fixed criteria, the same in every region	
Base flow index	Reference curve from 10 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high	
Degree of regulation	Fixed criteria, the same in every region	
Bankfull width:depth ratio	Reference curve from 17 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high	
Residual pool depth	Reference curve from 12 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high	
Bankfull height above channel	Reference curve from 17 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high	
Sand + fines	Reference curve from 17 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk moderate	
Embeddedness	Reference curve from 12 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk moderate	
Sinuosity	Reference curve from 12 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high	
Bank angle	Reference curve from 12 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk moderate	
Relative bed stability	Reference curve from 17 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high	
Substrate size	Reference curve from 17 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk moderate	
Large wood volume	Reference curve from 14 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high	
Canopy density, mid-channel	Reference curve from 12 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high	
Riparian veg cover, woody	Reference curve from 14 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high	
Natural fish cover	Reference curve from 14 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high	
Mapped dams within one mile	Fixed criteria, the same in every region	

Metrics not scored

These metrics are not scored. A metric withheld for insufficient reference support has no curve. No pool of least-disturbed stations in this ecoregion or its parents supports it, no national donor pool was shown to transfer, no modeled expectation passed validation, and no published criterion applies.

Metric	Functions	Reason
EPT taxa richness	Community dynamics	Insufficient reference support
Shannon diversity (H')	Community dynamics	Insufficient reference support
Tolerant individuals (%) ~ HBI	Community dynamics	Insufficient reference support
Benthic total taxa richness	Population support	Insufficient reference support

Assessment status: Preliminary. Reviewed and approved as Preliminary.

Metric	Functions	Reason
Chlorophyll a	Light and thermal regime	Insufficient reference support
Specific conductivity	Water and soil quality	Insufficient reference support
Dissolved nitrogen	Nutrient cycling	Insufficient reference support
pH	Water and soil quality	Insufficient reference support
Total phosphorus	Nutrient cycling	Insufficient reference support
Turbidity	Water and soil quality	Insufficient reference support
Native fish taxa richness	Population support	Insufficient reference support

Scores are computed automatically from the assessment's reference curves. Confirm state/region applicability of the curve source.